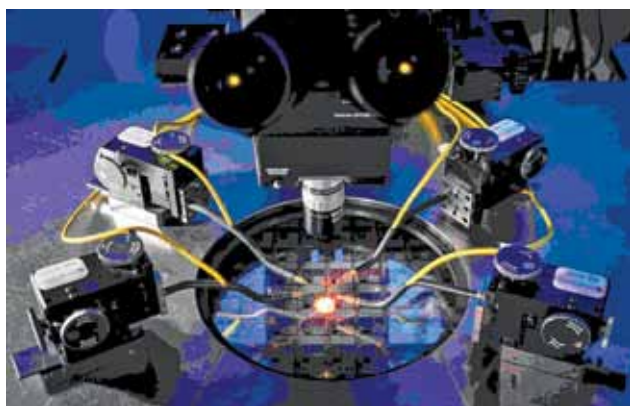


Radiation-resilient chip design toolkit

An open source radiation-resilient chip design toolkit at Fraunhofer IIS in Germany gives engineers a way to model how semiconductor components degrade under long-term



Fraunhofer researchers measure test structures on a semiconductor wafer (Credit: Fraunhofer IIS, Foto: BLEND 3 Frank Grätz)

radiation, helping improve reliability in space systems, medical imaging, and high-radiation industrial electronics. By simulating ageing effects on transistors, interconnects, and memory, designers can predict failure points and optimise redundancy early, reducing redesigns and silicon overhead while boosting resilience in extreme environments.

Next-gen biodegradable battery

A biodegradable battery by researchers at McGill University's Trottier Institute for Sustainability in Engineering and Design (Montreal, Quebec) that safely dissolves in water without heavy metals could power next-gen wearables, environmental sensors, and medical implants with minimal e-waste. The water-activated zinc battery delivers up to 1.2V and maintains performance across multiple discharge cycles, making it suitable for short-lifecycle devices where safe disposal and sustainability are critical. This advance supports greener consumer tech and transient electronics with lower environmental impact

Faster, energy-efficient data links for supercomputing and metaverse

A new 108Gb/s high-speed data receiver frontend boosts signal fidelity while cutting energy use, addressing key hurdles in hyperscale data centres, AI clusters, and networking gear where efficient, high-throughput links matter most. Researchers at Hanyang University, South Korea said that the multi-path architecture improves linearity with minimal power rise and compensates channel loss more precisely, promising faster, more energy-efficient data links for future 800G/1.6T Ethernet, supercomputing and metaverse infrastructure.

Multirate modelling for safer and reliable multi-sensor systems

A new multirate modeling technique tackles asynchronous data from mixed-rate sensors, improving autonomous navigation, robotics sensing, and industrial monitoring by unifying uneven sampling into a consistent system model. The algorithm by researchers at Kumamoto University, Japan, converts mismatched camera, lidar, and IMU streams into a time-invariant form, sharpening sensor fusion and reducing control errors, making real-world systems more reliable and safer in complex, multi-sensor applications

Needle-free glucose check

Researchers at MIT (Cambridge, Massachusetts) have developed a needle-free glucose-monitoring patch that enables painless, continuous blood-sugar tracking for diabetes management, remote patient monitoring, and wearable



A noninvasive blood-glucose monitoring method developed at MIT could spare diabetes patients from frequent finger pricks and potentially replace traditional monitoring devices (Credit: Christine Daniloff, MIT)

health devices. Using ultrasonic waves to extract interstitial fluid through the skin, the device delivers real-time glucose readings without pricking, reducing the risk of infection and improving user compliance. Its compact form factor allows integration with smartphones and smartwatches, paving the way for non-invasive, daily metabolic health tracking.

Dinner-plate-size AI chips reduce latency and improve efficiency

Dinner-plate-size AI chips developed by researchers at the University of Edinburgh show the potential to slash latency and boost inference speeds up to ~100x compared with GPU clusters, radically improving real-time AI tasks like reasoning assistants, fraud detection, scientific analytics, and other data-heavy workloads. By keeping large language models on a single wafer and optimising core-level computation, these processors reduce cross-chip communication delays and improve energy efficiency, offering a new hardware pathway for future AI systems.